YEAR 7 · MATHS UNIT 5 · SCIENCE UNIT 4 · HOMEWORK 12

None of the Seven

Part 1: Maths — 5.3 Drawing lines and quadrilaterals, and a review of 5.1 and 5.2
Part 2: Science — 4.1 Characteristics of living organisms and 4.2 Viruses

Name: _____ **Class:** _____

Date given: _____ **Date due:** Monday 16 November 2026

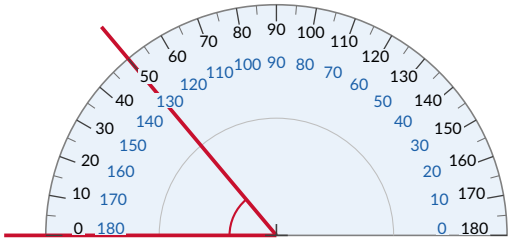
How to do this homework

- **A–B:** maths. **C–D:** science. **No calculator. You do not need a protractor**: a pencil and a ruler are enough.
- Write the **calculation** first, then the answer. **Give a reason** = write the angle fact after **because**.
- Answer in **full sentences** when it says so. *"Respiration"* is not a sentence. *"The gecko gets energy from its food by respiration"* is.

Section A · Maths 5.3 — Lines and Quadrilaterals

A1 Read the protractor

This is a picture of a protractor. You do not need your own.



(a) Look first. Is the red angle **acute** or **obtuse**?

_____

(b) The red line passes **two** numbers. Write them both.

_____ and _____

(c) What size is the red angle? _____

(d) How do you know? Finish the sentence.

"I count up from the 0 that …"

A2 The Workbook's words

Match each question to what it means. Write the **number** in the box. (a) is done for you.

(a) Draw the line through B perpendicular to AB. [4]

(b) Draw a perpendicular line from C to AB. [ ]

(c) Draw a line through P parallel to AB. [ ]

(d) Mark the point D. [ ]

(e) Measure the length of AD. [ ]

(f) How can you check that your answer is correct? [ ]

It means …

1 Use a ruler. Give the length to the nearest mm.

2 Start at C. Stop on AB. Meet AB at 90°.

3 It goes past P. It is always the same distance from AB.

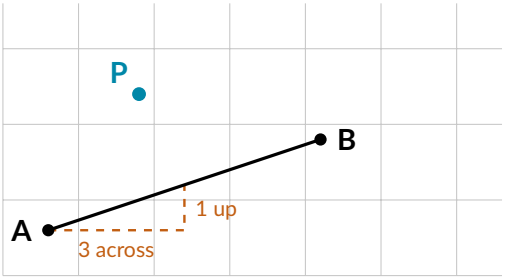
4 It goes past B, at 90° to AB.

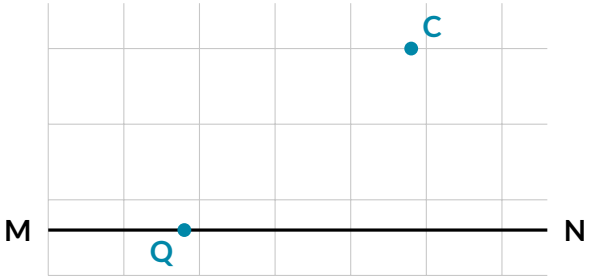
5 Use a fact, like "the four angles add up to 360°".

6 Make a small cross. Write D next to it.

A3 On the grid: parallel and perpendicular

Use a ruler. Count the squares — no protractor.





(a) Draw the line through **P** parallel to **AB**.
AB goes 3 squares across and 1 square up. Your line does the same.

(b) Draw the line through **Q** perpendicular to **MN**.

(c) Draw the perpendicular from **C** to **MN**. Mark the point **D** where it meets MN.

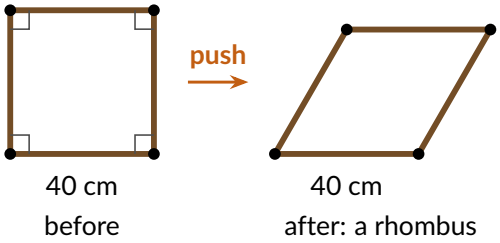
(d) How many squares long is **CD**? _____

(e) Mr Bowen walks from C to N. Is that longer or shorter than CD? Why?

Start: **"CD is the shortest way from C to MN because …"**

A4 A square needs right angles

Mr Bowen makes a square frame from four sticks. Each stick is 40 cm long. He puts **one** nail in each corner. Then he pushes it.



(a) Tick (✓) each box that is true.

Shape	4 equal sides	4 right angles
square		
rhombus		
rectangle		

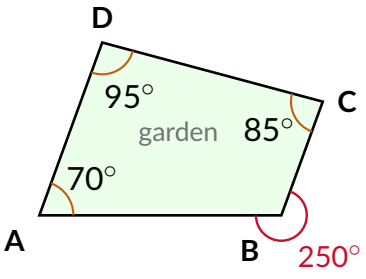
(b) Mr Bowen says: *"My frame still has four sides of 40 cm. So it is still a square."* Is he right? Answer in a full sentence.

Start: **"No, because a square also needs …"**

A5 Does it add up to 360°?

(a) Mr Bowen draws four quadrilaterals. He measures the four angles of each one. Add them up. Is the drawing **accurate**? Row 1 is done for you.

Drawing	The four angles	Add them up	Accurate?
1	85°, 95°, 100°, 80°	85 + 95 + 100 + 80 = 360°	yes
2	70°, 120°, 110°, 50°		yes · no
3	64°, 116°, 64°, 116°		yes · no
4	90°, 90°, 90°, 100°		yes · no



(b) This is Mr Bowen's garden. He measures the angle at **B** and reads **250°**.

(i) Did he measure the inside or the outside?

_____

(ii) What is the angle **inside** the garden at B?

Calculation: _____ **Answer:** _____

(iii) Check: do the four angles inside the garden add up to 360°? Write the calculation.

_____

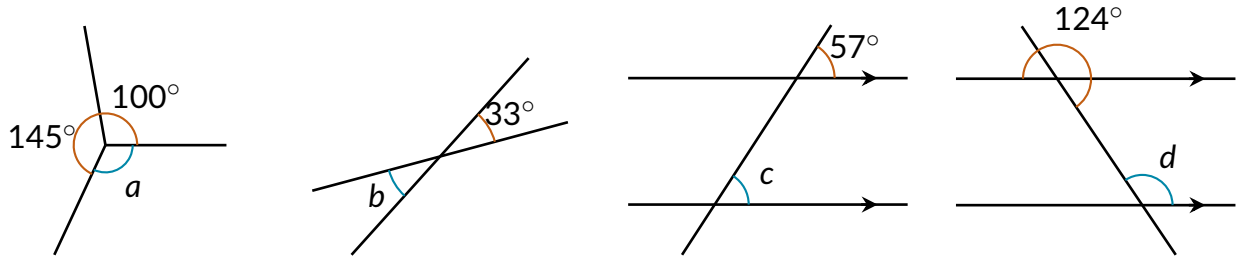
Section B · Review of 5.1 and 5.2 — Calculate, Do Not Measure

B1 Find the angle. Give a reason.

The reasons — copy the one you need after because

line: angles on a straight line add up to 180°
point: angles around a point add up to 360°
shape: angles in a quadrilateral add up to 360°
X: vertically opposite angles are equal
F: corresponding angles are equal
Z: alternate angles are equal

Arrows (>) on two lines mean they are **parallel**. Calculate each letter. Then give a reason.



(a) **(b)** **(c)** **(d)**

a = _____ because _____

b = _____ because _____

c = _____ because _____

d = _____ because _____

Section C · Science 4.1 — Characteristics of Living Organisms

C1 The seven, and one more

Match each word to its meaning. Write the **letter** in the box. The first one is done for you.

A a living thing
B getting bigger
C noticing changes around it
D getting rid of waste, such as carbon dioxide
E changing position or shape
F taking in or making food
G making more organisms of the same kind
H breaking down food inside cells to get energy

organism [A] movement [ ] respiration [ ] sensitivity [ ]

growth [ ] reproduction [ ] excretion [ ] nutrition [ ]

C2 The gecko on the wall

A gecko lives on the wall of Mr Bowen's classroom. Which of the seven is each clue? Row 1 is done for you.

Clue	Which of the seven?
1. The gecko runs up the wall.	movement
2. It sees a fly and turns its head.	
3. It eats the fly.	
4. Inside its cells, the fly is broken down to give energy.	
5. It breathes out carbon dioxide.	
6. It lays two eggs.	
7. A baby gecko is 3 cm long. A month later, it is 5 cm long.	
8. The gecko falls onto Mr Bowen's desk. Mr Bowen sees it and jumps. **Name two** of the seven that Mr Bowen does.	

C3 Breathing or respiration?

Tick (✓) **true** or **false**.

	True	False
1. Respiration is the same as breathing.		
2. Respiration happens inside cells.		
3. Breathing brings in the oxygen that respiration uses.		
4. A tree does not breathe in and out like us, so it does not do respiration.		
5. When you run, your cells do more respiration.		
6. In science, **sensitive** means "gets upset easily".		

Choose **one** false sentence. Write it again so that it is true.

C4 Alive, alive once, or never alive?

Word help

Alive: it is a living organism now. **Alive once:** it came from a living thing, but it is not alive now (a wooden desk was a tree). **Never alive:** it was never a living thing.

Write **A** (alive), **O** (alive once) or **N** (never alive) in each box.

a mango tree ☐ a wooden pencil ☐ a river ☐

a mango seed ☐ a candle flame ☐ a cotton T-shirt ☐

mould on old bread ☐ a glass bottle ☐ a sheet of paper ☐

Mr Bowen says: *"The river is alive. It moves, and it gets bigger in the rainy season."* Is he right? Answer in full sentences.

Start: **"No. A river moves and gets bigger, but it does not …"**

C5 The robot dog and the cactus

Remember from the lesson

A living organism does **all seven**. A non-living thing may do some, but **not all**.

Mr Bowen buys a robot dog. It walks around the room. It barks when someone comes in. When its battery is low, it goes to its charger. It never gets bigger. Mr Bowen says it loves him.

(a) Is the robot dog alive? Answer in full sentences. Name **two** of the seven it seems to do, and **two** it **cannot** do.

Start: **"The robot dog seems to do … But it is not alive because it cannot …"**

(b) A cactus does not walk, and it does not eat. But it **is** alive. Explain, in full sentences. Name **three** of the seven that a cactus does.

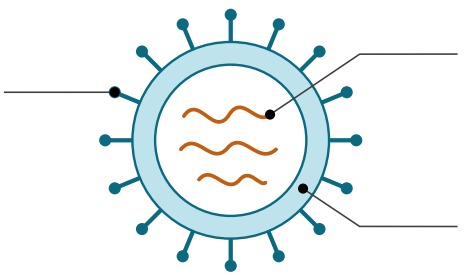
Start: **"A cactus makes its own food using light, so …"**

Section D · Science 4.2 — Viruses

D1 Label the virus

Label each part: **pegs** · **protein coat** · **RNA**. The grey words say what it does.

1 _____
sticks to a cell and helps the virus get inside



2 _____
instructions for making more viruses

3 _____
the outside of the virus

D2 A cell or a virus?

Tick (✓) the **cell**, the **virus**, or **both**.

	A cell in Mr Bowen's nose	A flu virus
1. It has a cell membrane, cytoplasm and a nucleus.		
2. It has a protein coat with pegs.		
3. You can see it with a school microscope.		
4. You can only see it with an electron microscope.		
5. It does respiration.		
6. You cannot see it with your eyes alone.		

D3 Follow the flu virus

Word help

replicate: make copies · **burst**: break open suddenly · **flu**: short for **influenza**

(a) A flu virus replicates. Put the steps in order: write **1** to **5**. Step 1 is done for you.

- [ ] The cell copies the RNA. It builds many new viruses.
- [1] The pegs stick to a cell in the nose.
- [ ] The cell bursts and dies. The new viruses float away.
- [ ] The virus gets inside the cell.
- [ ] The virus bursts open. Its RNA comes out.

(b) Mr Bowen says: *"A virus makes copies of itself, like a photocopier."* What is wrong with this?
Start: **"The virus does not make the copies. The …"**

(c) Why do you feel ill when you have flu?
Start: **"When the new viruses burst out, the cells …"**

D4 Mr Bowen's model virus

Mr Bowen makes a model of a flu virus. He uses a ball of green modelling clay, 20 push pins, and short pieces of red wool, pushed inside the clay.

(a) What is each part of the model?

In the model	In a real virus
the green clay ball	
the push pins	
the red wool	

(b) Give **one** way the model is **not** like a real virus.

Start: **"A real virus is …"**

D5 Is a virus alive?

(a) On its own, outside a cell, does a virus do each of the seven? Write **Y** (yes), **N** (no) or **S** (sort of).

movement ☐ respiration ☐ sensitivity ☐ growth ☐

reproduction ☐ excretion ☐ nutrition ☐

(b) Some scientists say a virus is alive. Others say it is not. What do **you** think? Write 5 sentences.

Start: **"I think a virus is / is not alive because …"**

Your answer must have: tick each one

- ☐ what **you** think, and **because**
- ☐ a reason for the **other** side
- ☐ the word **cells**
- ☐ the word **replicate**
- ☐ one of the **seven**, by name
- ☐ full sentences

Answer Key — Teacher Copy

A1 Read the protractor

(a) acute (b) 50 and 130 (c) **50°**
(d) *I count up from the 0 that sits on the line (the outside scale). The angle is acute, so it is less than 90°.*

130° is the deck's vote and the commonest error: the student read the scale whose 0 is on the *other* side. The estimate in (a) is there to catch it — an acute angle cannot be 130°. A student who writes "acute" and then 130 has not connected the two: point at their own answer to (a).

A2 The Workbook's words

(a) 4 (b) 2 (c) 3 (d) 6 (e) 1 (f) 5

(a) and (b) are the pair that matters: *through* B means the line passes B; *from* C *to* AB means start at C and stop on AB. A student who swaps 2 and 4 will draw A3(b) and A3(c) the wrong way round too.

A3 On the grid

(a) Any line through P that goes 3 across and 1 up, e.g. through (0, 3), P (3, 4), (6, 5) and (9, 6), counting squares from the bottom left.
(b) The vertical line through Q. (c) The vertical line from C down to MN; D is 4 squares below C. (d) **4** squares.
(e) *CN is longer. CD is the shortest way from C to MN because it meets MN at 90°.*

(a) Check the gap: the two lines are the same number of squares apart at both ends. Expect a line that goes 1 across and 3 up (the steps swapped) — it crosses AB, so it cannot be parallel; and a flat line through P, which "looks parallel" only because AB is not steep.
(b) and (c) are the same drawing move and different English. A student whose line for (b) runs from Q up to C has read *through Q* as *from Q to C*. A line for (c) that ends at Q (not straight down) has the perpendicular wrong: ask what angle it makes with MN.
(e) CN is 5 squares (3 across, 4 down); nobody needs the number. The reason is the write panel: 90° and shortest.

A4 A square needs right angles

(a) square: ✓ ✓ rhombus: ✓ (sides only) rectangle: ✓ (right angles only)
(b) *No, because a square also needs four right angles. The corners turned, so the frame is now a rhombus.*

The rectangle's "4 equal sides" box: accept it left blank. A student who ticks it is thinking of a square, which is a special rectangle; that is fine to discuss, not to mark wrong.
(b) A student who agrees with Mr Bowen has kept only half the definition. The deck's fix is the rulers: four equal sides, pushed, still four equal sides.

A5 Does it add up to 360°?

2: 70+120+110+50 = 350°, **no** 3: 64+116+64+116 = 360°, **yes** 4: 90+90+90+100 = 370°, **no**
(b)(i) the outside (ii) 360 − 250 = **110°** (iii) 110 + 70 + 85 + 95 = 360°: yes.

Drawing 4 is three right angles and one that is not: a student who says "yes" saw a rectangle and did not add.
(b) is the deck's table (270°) with an angle that is not a right angle. Expect 250 − 180 = 70 from a student who reaches for 180: the red arc goes all the way round the outside, so it is 360. (iii) is the payoff: the 110 they found makes the four add up, which is how they know (ii) is right.

B1 Find the angle. Give a reason.

a = 115° because angles around a point add up to 360° (360 − 100 − 145).
b = 33° because vertically opposite angles are equal.
c = 57° because corresponding angles are equal.
d = 124° because alternate angles are equal.

Accept the reason in the student's own words if the fact is right ("they make a Z" is fine for *d*). A number with no reason is half done.
(a) Expect 180 − 245, which cannot be done, from anyone who uses 180 for everything: ask "180 or 360?"
(b) Expect 147° from a student who took the angle *next to* 33°.
(d) is obtuse on purpose. Expect 56° (180 − 124) from a student who believes alternate angles are always the small ones.
The straight line and the quadrilateral are not repeated here: A5 already uses 360° in a quadrilateral twice.

C1 The seven, and one more

movement **E** respiration **H** sensitivity **C** growth **B** reproduction **G** excretion **D** nutrition **F**

These are the four write panels word for word, so a student with a complete notebook can check every one. Respiration and excretion are the pair most often swapped (both involve gases).

C2 The gecko on the wall

2 sensitivity 3 nutrition 4 respiration 5 excretion 6 reproduction 7 growth 8 sensitivity and movement

4 is the one to watch: "breathing" is not one of the seven, and the clue says *inside its cells*. 5 is excretion, not respiration — the waste leaving. 8 is deadpan; Mr Bowen noticed a change (sensitivity) and jumped (movement). Accept respiration as a third.

C3 Breathing or respiration?

1 false 2 true 3 true 4 false 5 true 6 false
Corrected, any one: *Respiration is not the same as breathing. / A tree does respiration in its cells, even though it does not breathe like us. / In science, sensitive means noticing changes around it.*

Expect most of the class to get 1 wrong: it is the deck's vote (slide 7), and French has the same trap. 4 is the book's own sentence turned round: "Plants don't breathe or have hearts, yet they are alive." A correction that just adds "not" is fine if the sentence is now true.

C4 Alive, alive once, or never alive?

A: mango tree, mango seed, mould **O:** wooden pencil, cotton T-shirt, sheet of paper **N:** river, candle flame, glass bottle
No. A river moves and gets bigger, but it does not eat, respire or reproduce. A living organism does all seven, so a river is not alive.

The traps, in the order the game taught them: the seed is alive (it is resting, like the dry beans); wood, cotton and paper were alive once (the wooden chair); the candle flame moves, grows and uses oxygen and is still never alive (fire); the mould is alive. A student who writes N for the pencil has the deck's starter error — it was a tree. The river is the cloud again: it moves and grows. Accept any two of the five it does not do.

C5 The robot dog and the cactus

(a) *The robot dog seems to do movement and sensitivity (it notices someone come in). But it is not alive because it cannot grow or reproduce. It does not have cells, so it cannot do respiration. A living organism does all seven.*
(b) *A cactus makes its own food using light, so it does nutrition. It grows, it makes seeds (reproduction), and it does respiration in its cells. It moves slowly, towards the light. It does all seven, so it is alive.*

The two halves are the same rule from opposite ends: the robot dog seems to do a lot and is not alive; the cactus seems to do little and is. (a) Seems to do: movement, sensitivity, and nutrition "sort of" (it takes in electricity — the deck gave the car's fuel a gold wave). Cannot do: any two of growth, reproduction, respiration, excretion. "It loves him" is not one of the seven.
(b): any three of the seven, each with something the cactus does.

D1 Label the virus

1 pegs 2 RNA 3 protein coat

Pegs and coat are the pair to check: label 1 lands on a peg's head, label 3 on the ring. A student who swaps them is reading "coat" as the outermost thing — point out the pegs stick *out of* the coat.

D2 A cell or a virus?

Cell: 1, 3, 5, 6 Virus: 2, 4, 6 (6 is both)

5 is 4.1 again: no cells, so no respiration. 6 catches students who think a cell is big enough to see; most body cells are about 0.02 mm. A tick for the virus in row 3 means the student has not met the two microscopes: slide 5.

D3 Follow the flu virus

(a) 4, 1, 5, 2, 3 (stick, get in, open, copy, burst out)
(b) *The virus does not make the copies. The cell makes them, and then the cell dies.*
(c) *When the new viruses burst out, the cells die. That makes you feel ill.*

(a) The usual slip is "bursts open" (the virus, 3) against "bursts and dies" (the cell, 5): read who bursts.
(b) is the vote on slide 9 that most of the class got wrong before Be the Cell. A student who writes "the virus makes them inside the cell" has not changed their mind: ask who did the copying in the game.

D4 Mr Bowen's model virus

(a) clay ball: protein coat push pins: pegs red wool: RNA
(b) Any one: *A real virus is much smaller* (about 0.0001 mm) / *is not coloured* (the colours in the photos were added) / *is not made of clay*.

"It is not alive" is not a difference: the model is not alive either. A good answer names one thing and says how the real virus is different.

D5 Is a virus alive?

(a) movement N, respiration N, sensitivity N, growth N, reproduction **S**, excretion N, nutrition N (the deck's slide 16).
(b) Check the tick list, in order. Model: *I think a virus is not alive because it is not made of cells. On its own, it does none of the seven: it does not do respiration or nutrition. But some scientists say it is alive, because it has RNA and it can replicate. It can only replicate inside a living cell. So I think it is not alive.*

Either answer is right if it is argued: the book leaves it open, and so did the deck. Accept N for reproduction with a reason ("the cell does the work"). The weakest answers give only one side; the tick list asks for a reason for the *other* side, so a "not alive" answer needs a "but" sentence.